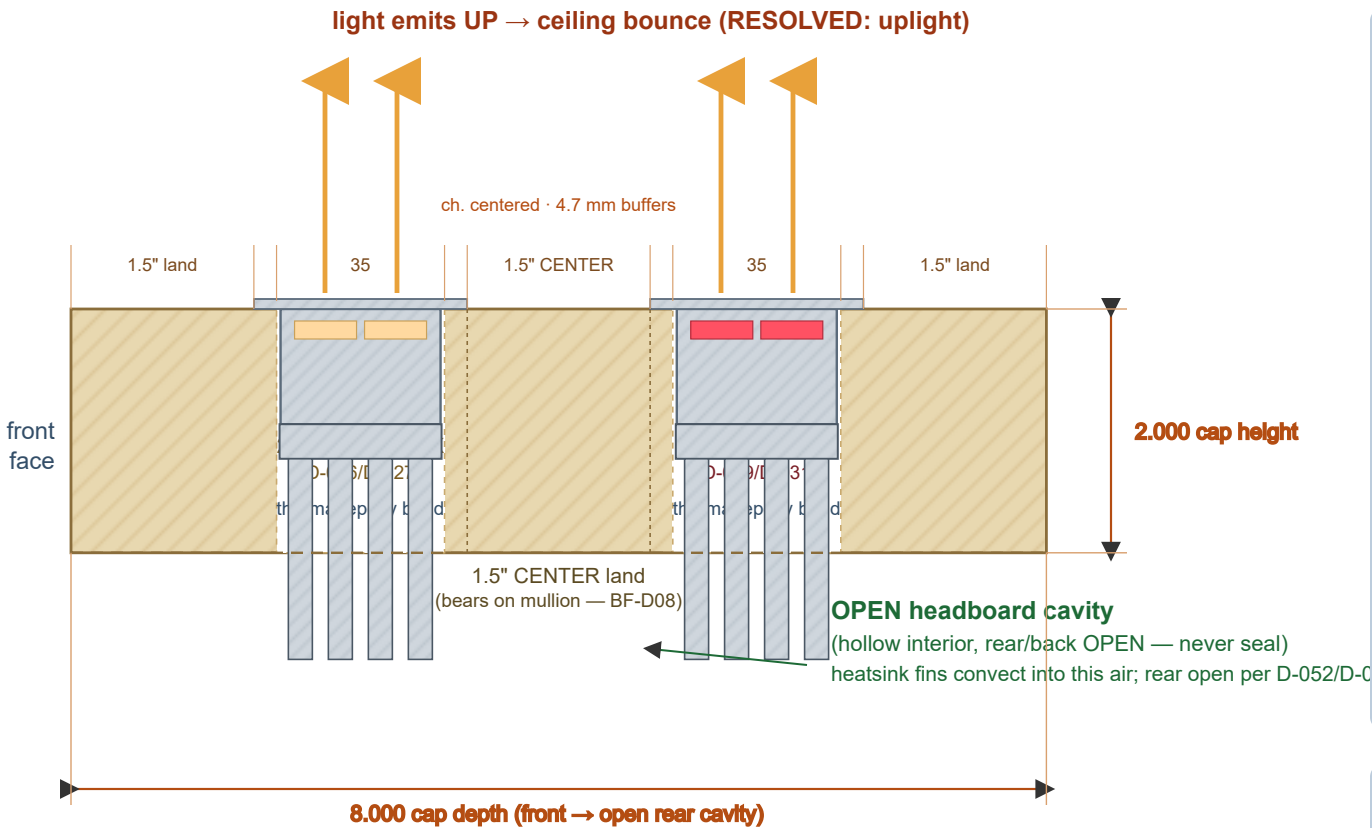


BF-D01 — TOP-CAP LED CHANNEL — CROSS-SECTION (sheet 1 of 2)

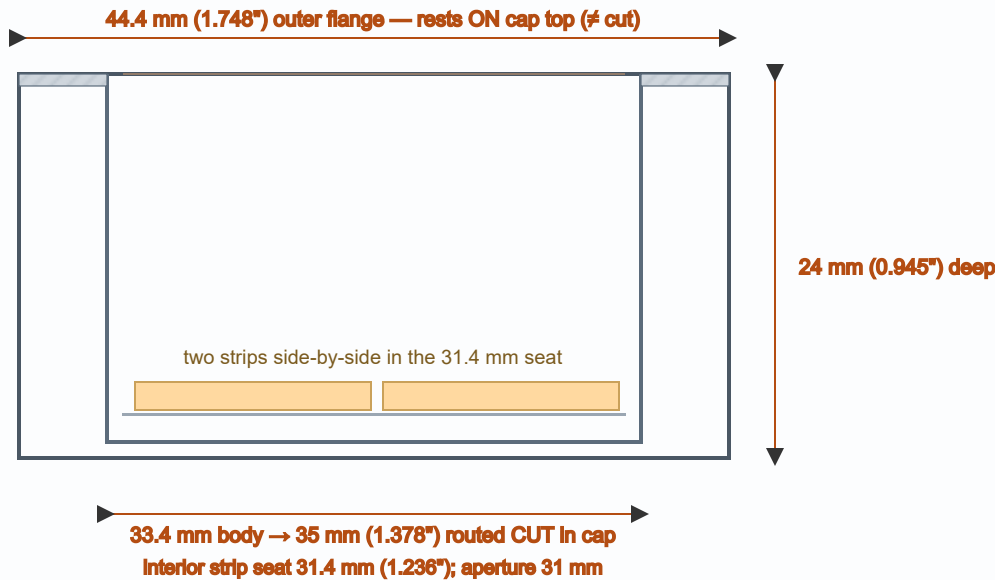
Solid Ash Heirloom Bed — section cut across the 124" run — cap 8" deep × 2" tall — TWO identical Vario30-07 channels, mouths UP (uplight / ceiling-bounce), 1551AS heatsinks bonded to each channel back, fins into the OPEN headboard cavity.

VIEW: looking along the 130" run (cut across it). FRONT of headboard = left → REAR (open) = right. UP = toward ceiling.



DETAIL — Vario30-07 channel (BOTH top-rail channels, identical)

From vendor drawing (LED-Hardware-Reference.md). Body recessed in a 35 mm routed THROUGH-slot; the 44.4 mm flange rests



DESIGN INTENT

- Two IDENTICAL Vario30-07 channels in 35 mm through-cuts, centered in the gaps between THREE 1.5" lands (front / center / rear). Each holds TWO strips. Warm/front = 2× Auxmer 3000K; red/rear = 2× 660 nm.
- Channel mouths face UP → uplight / ceiling-bounce.
- A HeatsinkUSA 1551AS (black-anodized 6063-T5) bonds with thermal epoxy to EACH channel back; fins drop into the OPEN hollow headboard cavity for convection. Covers full length except a 1–3" center power gap.

NOTES

- Rear/underside cavity is OPEN (headboard rear open, no continuous back panel) — mandatory thermal + service access; NEVER seal (LEDGER:D-052/D-060).
- Heatsink 1551AS: 33.8 mm W × 49 mm tall (41.8 mm fins + 7.2 mm base), 4 fins, ~2.76 °C/W per 3".
- 1551AS width 33.8 mm ≈ Vario30-07 33.4 mm back → near-perfect bond face (D-054).
- 44.4 mm flange rests ON the cap top; cap held down by 3 vertical insert draw-bolts per mullion (1 per land), D-052.
- 1/4" radius at every channel corner (plunge-router reality). All heights measured from the floor.

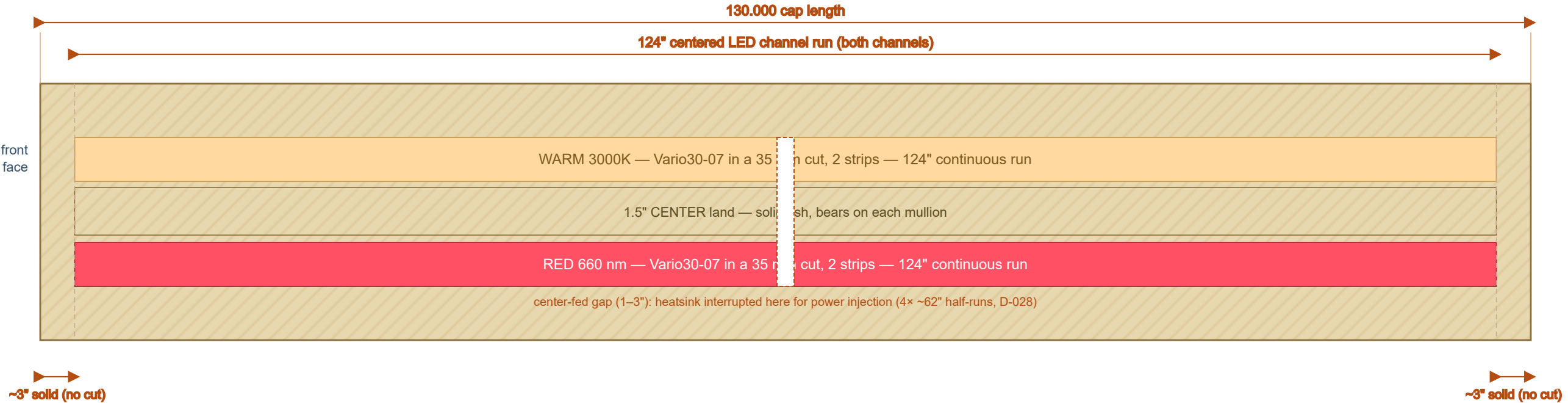
OPEN ITEMS

- Exact land/cut setback within the 8" depth → builder routing drawing (D-061): three 1.5" lands, 35 mm cuts centered.
- Slot depth / rabbet depth / channel retention (inserts vs set-screw) — confirm w/ real Vario sample (D-061).
- Bench-confirm < 50–55 °C at full power; active-fan failsafe TBD (D-051/D-054).

PROJECT	Solid Ash Heirloom Bed
PART / CODE	Top-cap LED channel — BF-D01
STATUS / DATE	CURRENT · 2026-06-05
DATE / SHEET	2026-06-05 — 1 of 2 — SECTION
SCALE	enlarged section — not to scale — DO NOT SCALE
MATERIAL	Solid ash; quartersawn (deliver RAW/unfinished)
FINISH	Black penetrating dye + 2K matte poly (owner-applied)
SOURCE OF TRUTH	LEDGER D-027 / D-031 / D-033 / D-052 / D-054
REVISIONS	history in git + CHANGELOG.md
TOLERANCE	±1/32 in unless noted otherwise

BF-D01 — TOP-CAP LED CHANNEL — PLAN / TOP VIEW (sheet 2 of 2)

Solid Ash Heirloom Bed — cutout positions along the ONE-PIECE 130" cap — 124" centered run routed in, ~3" solid ash at each end (NO separate blocks) — both channels Vario30-07 (depth scale exaggerated).



LEGEND — part colors

- One-piece solid-ash cap (the 3 lands incl. the center land + ~3" ends are all the same)
- Warm 3000K channel (Vario30-07, 2 strips)
- Red 660 nm channel (Vario30-07, 2 strips)
- Center power-injection gap (1–3")

Both channels = identical Vario30-07 (D-027/D-031). Smart-In16 is used only for accent runs, not for main runs.

NOTES

- Two 35 mm through-cuts, 124" long, centered in the gaps between three 1.5" lands (front/center/rear).
- The cap is ONE 130" ash board; the 124" run is routed IN — no separate end blocks.
- Each channel = Vario30-07 holding TWO strips side-by-side (warm: Auxmer; red: SuperLightingLED).
- 1551AS heatsink bonded to each channel back along the full run (less the center gap) → open cavity.
- Light emits UP (uplight / ceiling-bounce). See sheet 1 for the depth section + heatsink projection.
- All heights measured from the floor. DO NOT SCALE — read written dimensions.

OPEN ITEMS

- How the 124" run crosses the headboard support structure → engineering (D-060 / D-061).
- Channel retention method & slot/rabbet depths (D-061).
- Heatsink piece breaks & fan failsafe (D-051 / D-054).

CURRENT (2026-06-05): two Vario30-07 channels, 35 mm cuts centered in the gaps between three 1.5" lands (front/center/rear); 44.4 mm flange rests on top; uplight; ONE-PIECE cap (channels routed in); open rear cavity + 1551AS heatsink into it; cap held by 3 vertical insert bolts per mullion. Revision history in git + CHANGELOG.md.

PART / CODE	Top-cap LED channel — BF-D01
STATUS / DATE	CURRENT · 2026-06-05
DATE / SHEET	2026-06-05 — 2 of 2 — PLAN
SCALE	not to scale (depth exaggerated) — DO NOT SCALE
SOURCE OF TRUTH	LEDGER D-027 / D-031 / D-033 / D-052 / D-054
REVISIONS	history in git + CHANGELOG.md